ELEC70009
EE9-CS2-2

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE/EIE PART IV: MEng and ACGI

DISCRETE-EVENT SYSTEMS

Wednesday, 7 May 2025, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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1. A robot operates by assembling two different types of pieces (type 1 and 2, respectively) into a single object. Arrivals of pieces at the robot of type $i = 1,2$ are modeled as a_i events. Each type of piece can be stored in a dedicated buffer of up to 2 units at the robot. Both buffers start as empty. When pieces of type 1 and 2 are both available in the buffers, a processing event p can take place, and, as a result, new slots of type 1 and 2 are made available at the buffers while an object is produced by the robot.

   a) Model each buffer by means of Finite Deterministic Automaton G_i, $i = 1,2$, keeping track of available slots as a result of a_i and p events. [4]

   b) Model the operation of the assembly robot by means of a single Finite Deterministic Automaton G, with alphabet $\{a_1, a_2, p\}$. [4]

   c) Assume that the sensor detecting arrival of pieces cannot detect their type, (while the pieces are still automatically assigned to their correct buffer). Model this situation by means of a Finite Non-deterministic Automaton G_N. [4]

   d) Compute the observer automaton $\text{Obs}(G_N)$. What is the meaning of this automaton in this particular example? [6]

   e) What are the possible states reached after $w = aaaad$ is detected? What are the possible states reached after $w = aaada$ is detected? [2]

2. A machine M produces two types of products, A and B, respectively. It operates by processing two types of pieces, type 1 and 2, respectively. Arrivals of pieces of type i at the machine are modeled as a_i events. Each type of piece can be stored in a dedicated buffer of up to 2 units at the machine. Product A only needs a single piece of type 1. Each time product A is produced event p_A occurs and one slot from the buffer of type 1 is freed up. Product B needs both a piece of type 1 and type 2. Each time product B is produced event p_B occurs and one slot from the buffer of type 2 becomes available.

a) Model individual buffers by means of FDA G_i, $i = 1,2$, keeping track of the number of available slots as a result of a_1, a_2, p_A, p_B events, respectively. [3]

b) Compute the parallel composition $G = G_1||G_2$. [3]

c) The machine should operate according to a strict event alternation policy between p_A and p_B (so as to ensure that the number of A and B products produced is always equal, or at most different by one). Model this partial specification through a suitable automaton G_K, assuming that p_A is the first event to occur. [3]

d) Assuming $E_{uc} = \{a_1, a_2\}$ is the set of uncontrollable events, find out if the specification in item c) is controllable. (Hint: compute the parallel composition $G||G_K$). [4]

e) Assume next that $E_{uc} = \{p_B, a_1\}$. Is the partial specification described in item c) controllable? (Justify your answer). [2]

f) Find an admissible supervisor S such that the closed-loop language $\mathscr{L}(S/G)$ implements the maximal controllable sublanguage of the considered specification for $E_{uc} = \{p_B, a_1\}$. [5]

3. Consider the following network of 5 chemical reactions:

$$R_1 : A + B \rightarrow AB$$

$$R_2 : B + C \rightarrow BC$$

$$R_3 : C + AB \rightarrow ABC$$

$$R_4 : A + BC \rightarrow ABC$$

$$R_5 : ABC \rightarrow A + B + C$$

where A, B, C, AB, BC, ABC denote chemical species and R_i, $i = 1 \ldots 5$ are the 5 chemical reactions taking place among them. When reaction i occurs, the corresponding reactants (species on the left-hand side of the arrow) decrease by one unity, while the corresponding products (species on the right-hand side) increase by one unit.

a) Model the chemical reaction network as a Petri Net N, by associating places with chemical species and reactions to transitions. Sketch its diagram. [4]

b) Compute the incidence matrix C associated to the Petri Net. [2]

c) Find, if any, the P-semiflows of minimal support for the Petri Net. [4]

d) Find the T-semiflows of minimal support for the Petri Net. [4]

e) Consider next the situation in which initial marking M_0 is 1 molecule of A, 2 molecules of B and 1 molecule of C (and 0 molecules of all other compounds. Sketch the reachable set for this choice of initial marking. [3]

f) Comment on the reversibility of the network (N, M_0). Which compounds do you expect to find as time goes to infinity, assuming random collisions trigger reactions (i.e. a Markovian stochastic process). [3]

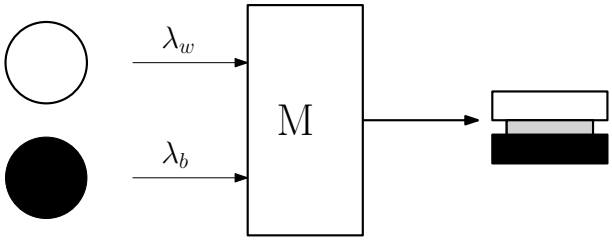


Figure 4.1 Cookie making machine

4. A cookie making machine M, uses cream to stick together with two round biscuits of different colour (white and black) (see Fig. 4.1). Arrival times of the biscuits at the machine are exponentially distributed with rates λ_w and λ_b, respectively. The machine does not have buffers, and while a biscuit of a certain colour is in, arrivals of further biscuits of that same colour do not occur (or are ignored and sent downstream to a different machine, which we are not modeling).

   a) Model the arrivals of biscuits at the machine for the production of a single cookie by means of a Markov chain. Sketch its transition diagram. [5]

   b) Write the Kolmogorov equation for the evolution of probabilities of the Markov chain described in item a). [5]

   c) What is the average time needed to produce a cookie assuming that the sticking phase has negligible duration with respect to the process of arrival of the biscuits? [6]

   d) How would this change if the machine operates by sticking together biscuits of arbitrary colours (viz. randomly producing cookies also with two whites or two black parts). Assume that it allows arrivals of white parts also when a white biscuit is already in or of black biscuits when a black one is already in. [4]